

### Question 1. Express the interval as inequalities and graph it

The interval is  $(-1, 1)$ . Parentheses mean the endpoints are not included. Therefore  $w$  must be greater than  $-1$  and less than  $1$ .

$$(-1, 1) \text{ means } -1 < w < 1$$

For the number-line graph, place open circles at  $-1$  and  $1$ , then shade the segment between them.

**Final answer:**  $-1 < w < 1$ ; graph with open circles at  $-1$  and  $1$ , shaded between.

### Question 2. Evaluate each expression

A power of  $1/3$  means cube root. Cube roots preserve the sign of negative numbers.

$$(a) 64^{1/3} = \text{cube root of } 64 = 4$$

**Final answer:** (a) 4

$$(b) (-27)^{1/3} = \text{cube root of } -27 = -3$$

**Final answer:** (b) -3

$$(c) -(1/27)^{1/3} = -(\text{cube root of } 1/27) = -1/3$$

**Final answer:** (c)  $-1/3$

**Teacher note:** A common mistake is to divide by 3 instead of taking a cube root. For example,  $64^{1/3}$  is 4, not  $64/3$ .

### Question 3. Multiply and simplify

Distribute each term in the first factor across the trinomial.

$$\begin{aligned} & (6x - 9)(x^2 - x + 1) \\ &= 6x(x^2 - x + 1) - 9(x^2 - x + 1) \\ &= 6x^3 - 6x^2 + 6x - 9x^2 + 9x - 9 \\ &= 6x^3 - 15x^2 + 15x - 9 \end{aligned}$$

**Final answer:**  $6x^3 - 15x^2 + 15x - 9$

### Question 4. Multiply/divide rational expressions and simplify

Factor the differences of squares first. Then cancel only common factors, never terms inside sums.

$$\begin{aligned} & [(x^2 - 81)/(x^2 - 25)] * [(x + 5)/(x + 9)] \\ &= [((x - 9)(x + 9))/((x - 5)(x + 5))] * [(x + 5)/(x + 9)] \\ &= (x - 9)/(x - 5) \end{aligned}$$

**Final answer:**  $(x - 9)/(x - 5)$

**Teacher note:** Original restrictions:  $x$  cannot be  $-9$ ,  $-5$ , or  $5$  because these values make an original denominator zero.

### Question 5. Solve the rational equation

Clear denominators using the least common denominator  $x^2(x - 9)$ .

$$\begin{aligned} & 9/(x - 9) - 18/x^2 = 0 \\ & \text{Multiply by } x^2(x - 9): \\ & 9x^2 - 18(x - 9) = 0 \\ & 9x^2 - 18x + 162 = 0 \\ & \text{Divide by 9:} \\ & x^2 - 2x + 18 = 0 \end{aligned}$$

Check whether the quadratic has real roots using the discriminant.

$$\begin{aligned}\text{Discriminant} &= b^2 - 4ac \\ &= (-2)^2 - 4(1)(18) \\ &= 4 - 72 \\ &= -68\end{aligned}$$

The discriminant is negative, so the quadratic has no real roots.

**Final answer: NO REAL SOLUTION**

## Question 6. Solve the nonlinear inequality

Find where each factor is zero, then make a sign chart.

$$\begin{aligned}x(2x + 5) &\geq 0 \\ \text{Critical points:} \\ x &= 0 \\ 2x + 5 &= 0 \rightarrow x = -5/2\end{aligned}$$

The critical points split the number line into three intervals:  $(-\infty, -5/2)$ ,  $(-5/2, 0)$ , and  $(0, \infty)$ . Test one value in each interval.

$$\begin{aligned}\text{For } x = -3: x(2x + 5) &= (-3)(-1) > 0 \\ \text{For } x = -1: x(2x + 5) &= (-1)(3) < 0 \\ \text{For } x = 1: x(2x + 5) &= (1)(7) > 0\end{aligned}$$

Because the inequality is  $\geq 0$ , include the positive intervals and include the endpoints where the product is zero.

**Final answer:  $(-\infty, -5/2] \cup [0, \infty)$**

Graph: closed circles at  $-5/2$  and  $0$ ; shade left of  $-5/2$  and right of  $0$ .

## Question 7. Find the equation of the line

Use slope-intercept form  $y = mx + b$ , where  $m$  is the slope and  $b$  is the y-intercept.

$$\begin{aligned}m &= 6/7, \quad b = 5 \\ y &= mx + b \\ y &= (6/7)x + 5\end{aligned}$$

**Final answer:  $y = (6/7)x + 5$**

## Question 8. Net change and average rate of change

The function is  $f(z) = 5 - 4z^2$ , from  $z = -2$  to  $z = 0$ .

$$\begin{aligned}f(-2) &= 5 - 4(-2)^2 \\ &= 5 - 4(4) \\ &= 5 - 16 \\ &= -11 \\ f(0) &= 5 - 4(0)^2 = 5\end{aligned}$$

Net change means final function value minus initial function value.

$$\begin{aligned}\text{Net change} &= f(0) - f(-2) \\ &= 5 - (-11) \\ &= 16\end{aligned}$$

Average rate of change is net change divided by the change in input.

$$\begin{aligned}\text{Average rate of change} &= [f(0) - f(-2)]/[0 - (-2)] \\ &= 16/2 \\ &= 8\end{aligned}$$

**Final answer: Net change = 16; average rate of change = 8**

### Question 9. Graph the quadratic and choose the most appropriate viewing window

The function is  $g(x) = x^2 - x - 20$ . To graph it accurately, identify intercepts, vertex, and opening direction.

$$\begin{aligned}g(x) &= x^2 - x - 20 \\ &= (x - 5)(x + 4)\end{aligned}$$

Therefore the x-intercepts are  $x = 5$  and  $x = -4$ . Since the coefficient of  $x^2$  is positive, the parabola opens upward.

$$\begin{aligned}\text{Vertex x-coordinate} &= -b/(2a) \\ \text{Here } a &= 1 \text{ and } b = -1, \text{ so} \\ x &= -(-1)/(2*1) = 1/2\end{aligned}$$

$$\begin{aligned}g(1/2) &= (1/2)^2 - 1/2 - 20 \\ &= 1/4 - 1/2 - 20 \\ &= -20.25\end{aligned}$$

So the vertex is approximately  $(0.5, -20.25)$ , and the graph crosses the x-axis at  $-4$  and  $5$ .

A good viewing window should include both x-intercepts and the vertex. The window  $[-7, 7]$  by  $[-25, 20]$  shows these features clearly without compressing the graph too much.

**Final answer: Most appropriate window:  $[-7, 7]$  by  $[-25, 20]$**

**Teacher note:** In the smaller  $[-2, 2]$  by  $[-5, 5]$  window, the graph is mostly below the visible y-range. In the very large  $[-10, 10]$  by  $[-100, 100]$  window, the graph is visible but vertically compressed.

### Question 10. Match each parent function with its graph

Use the standard shapes of the four parent functions.

$$\begin{aligned}f(x) &= x^2 \rightarrow \text{U-shaped parabola} \rightarrow \text{Graph IV} \\ f(x) &= x^3 \rightarrow \text{S-shaped cubic} \rightarrow \text{Graph II} \\ f(x) &= \sqrt{x} \rightarrow \text{starts at origin, right} \rightarrow \text{Graph I} \\ f(x) &= |x| \rightarrow \text{V-shaped graph} \rightarrow \text{Graph III}\end{aligned}$$

**Final answer: (a) IV, (b) II, (c) I, (d) III**

### Question 11. Factor the polynomial and find the zeros

Start by factoring out the common factor  $x$ .

$$\begin{aligned}P(x) &= x^3 + x^2 - 20x \\ &= x(x^2 + x - 20) \\ &= x(x + 5)(x - 4)\end{aligned}$$

Set each factor equal to zero.

$$\begin{aligned}x &= 0 \\ x + 5 &= 0 \rightarrow x = -5 \\ x - 4 &= 0 \rightarrow x = 4\end{aligned}$$

**Final answer: Zeros:  $-5, 0, 4$**

The graph is a positive-leading cubic. Its left end goes down, its right end goes up, and it crosses the x-axis at -5, 0, and 4.

### Question 12. Continuously compounded investment

For continuous compounding, use  $A = Pe^{rt}$ , where  $P$  is principal,  $r$  is the annual rate as a decimal, and  $t$  is time in years.

$$\begin{aligned} P &= 1000 \\ r &= 6.5\% = 0.065 \\ A &= 1000e^{0.065t} \end{aligned}$$

$$\begin{aligned} \text{(a) } t &= 3: \\ A &= 1000e^{0.065 \cdot 3} \\ &= 1000e^{0.195} \\ &= 1215.31 \end{aligned}$$

**Final answer: After 3 years: \$1215.31**

$$\begin{aligned} \text{(b) } t &= 6: \\ A &= 1000e^{0.065 \cdot 6} \\ &= 1000e^{0.39} \\ &= 1476.98 \end{aligned}$$

**Final answer: After 6 years: \$1476.98**

$$\begin{aligned} \text{(c) } t &= 18: \\ A &= 1000e^{0.065 \cdot 18} \\ &= 1000e^{1.17} \\ &= 3221.99 \end{aligned}$$

**Final answer: After 18 years: \$3221.99**

### Question 13. Find $\sin(t)$ , $\cos(t)$ , and $\tan(t)$ from the terminal point

The terminal point is  $(-3/5, -4/5)$ . On the unit circle, the x-coordinate is  $\cos(t)$ , and the y-coordinate is  $\sin(t)$ .

$$\begin{aligned} \cos(t) &= x = -3/5 \\ \sin(t) &= y = -4/5 \end{aligned}$$

Tangent is sine divided by cosine, or y divided by x.

$$\begin{aligned} \tan(t) &= \sin(t)/\cos(t) \\ &= (-4/5)/(-3/5) \\ &= 4/3 \end{aligned}$$

**Final answer:  $\sin(t) = -4/5$ ;  $\cos(t) = -3/5$ ;  $\tan(t) = 4/3$**

**Teacher note:** Because the point is in Quadrant III, both sine and cosine are negative, so tangent is positive.

### Question 14. Find side $x$ in the right triangle

The triangle has a right angle and a 45-degree angle. Therefore the third angle is also 45 degrees, so this is a 45-45-90 triangle.

In a 45-45-90 triangle, the side ratio is leg : leg : hypotenuse = 1 : 1 :  $\sqrt{2}$ .

$$\begin{aligned} \text{Given leg} &= 34 \\ \text{Hypotenuse} &= 34\sqrt{2} \end{aligned}$$

**Final answer:  $x = 34\sqrt{2}$**

### Question 15. Verify the trigonometric identity

Start from the left-hand side and use reciprocal identities.

$$\begin{aligned} & [\cos(u)\sec(u)]/[\tan(u)] \\ &= \cos(u) * \sec(u) * [1/\tan(u)] \\ &= \cos(u) * [1/\cos(u)] * \cot(u) \\ &= 1 * \cot(u) \\ &= \cot(u) \end{aligned}$$

**Final answer:**  $[\cos(u)\sec(u)]/[\tan(u)] = \cot(u)$ , so the identity is verified.

### Question 16. Evaluate the dot product expression

Write each vector in component form first.

$$\begin{aligned} u &= 4\mathbf{i} + \mathbf{j} = (4, 1) \\ v &= \mathbf{i} - 3\mathbf{j} = (1, -3) \\ w &= 3\mathbf{i} + 2\mathbf{j} = (3, 2) \end{aligned}$$

Now compute the two dot products separately.

$$\begin{aligned} u \text{ dot } v &= 4(1) + 1(-3) \\ &= 4 - 3 \\ &= 1 \end{aligned}$$

$$\begin{aligned} u \text{ dot } w &= 4(3) + 1(2) \\ &= 12 + 2 \\ &= 14 \end{aligned}$$

$$(u \text{ dot } v)(u \text{ dot } w) = (1)(14) = 14$$

**Final answer:** 14

### Question 17. Write the sum without sigma notation

The notation tells us to substitute  $k = 6, 7, 8$ , and  $9$  into  $k(k + 3)$ , then add the terms.

$$\begin{aligned} & \text{sum from } k = 6 \text{ to } 9 \text{ of } k(k + 3) \\ &= 6(6 + 3) + 7(7 + 3) + 8(8 + 3) + 9(9 + 3) \\ &= 6(9) + 7(10) + 8(11) + 9(12) \end{aligned}$$

If a numerical value is requested, continue simplifying.

$$\begin{aligned} &= 54 + 70 + 88 + 108 \\ &= 320 \end{aligned}$$

**Final answer:** Without sigma notation:  $6(6 + 3) + 7(7 + 3) + 8(8 + 3) + 9(9 + 3)$ . Simplified value: 320.

## Final Answer Key

| Q  | Answer                                                |
|----|-------------------------------------------------------|
| 1  | $-1 < w < 1$                                          |
| 2  | 4; -3; $-1/3$                                         |
| 3  | $6x^3 - 15x^2 + 15x - 9$                              |
| 4  | $(x - 9)/(x - 5)$                                     |
| 5  | NO REAL SOLUTION                                      |
| 6  | $(-\infty, -5/2] \cup [0, \infty)$                    |
| 7  | $y = (6/7)x + 5$                                      |
| 8  | Net change = 16; average rate = 8                     |
| 9  | Best window: $[-7, 7]$ by $[-25, 20]$                 |
| 10 | IV, II, I, III                                        |
| 11 | Zeros: -5, 0, 4                                       |
| 12 | \$1215.31; \$1476.98; \$3221.99                       |
| 13 | $\sin(t) = -4/5$ ; $\cos(t) = -3/5$ ; $\tan(t) = 4/3$ |
| 14 | $x = 34\sqrt{2}$                                      |
| 15 | $\cot(u)$                                             |
| 16 | 14                                                    |
| 17 | $6(6+3)+7(7+3)+8(8+3)+9(9+3)$ ; value 320             |